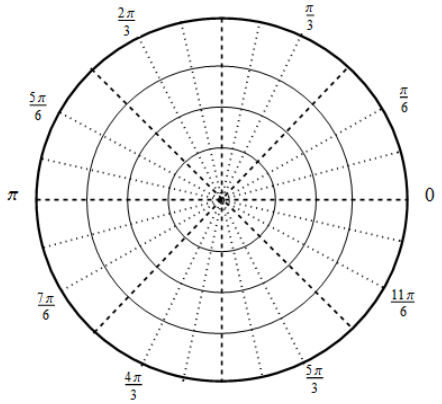
MATH 152 **QUIZ 10** NAME _____

Mrs. Bonny Tighe 25 points

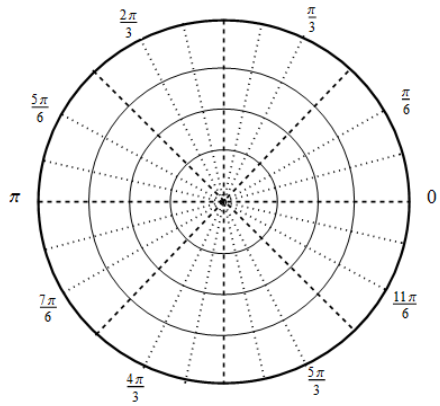
10.3 TA_____ Wed 5/8/2024

1. Find the equation of the tangent line to the curve $r = 1 - \sin 2\theta$ when $\theta = \pi/3$

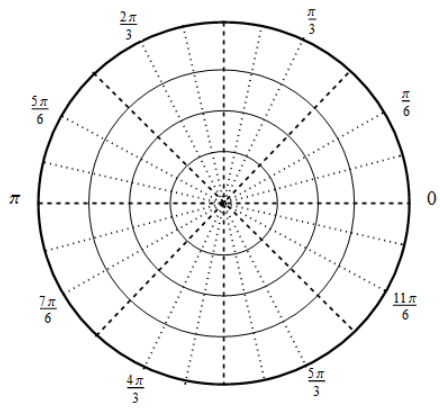
2. Sketch the graph. $r = 2 + 2\cos 2\theta$.



3. Sketch the graph $r = 2 - 3\sin\theta$



4. Sketch the graph $r = 2 + 2\cos\theta$



5. a) Find the Cartesian coordinates for the polar coordinate $(-2, 5\pi/6)$

b) Find polar coordinates for the Cartesian coordinate (-2, 2) where $r < 0, and\ \theta < 0$

c) Give two alternate versions on the polar point $(3, 5\pi/4)$

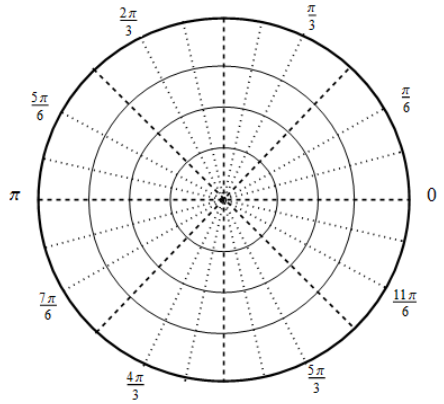
MATH 152 **QUIZ 10A** NAME _____

Mrs. Bonny Tighe 25 points

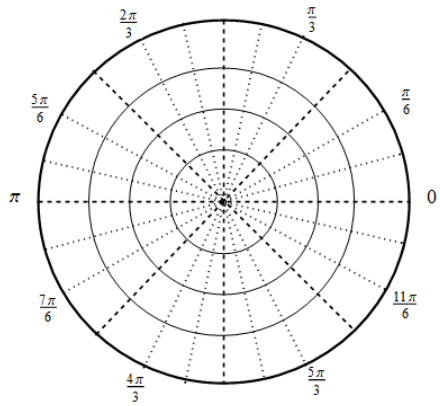
10.3 TA_____ Wed 5/8/2024

1. Find the equation of the tangent line to the curve $r = 2 - \cos 2\theta$ when $\theta = \pi/6$

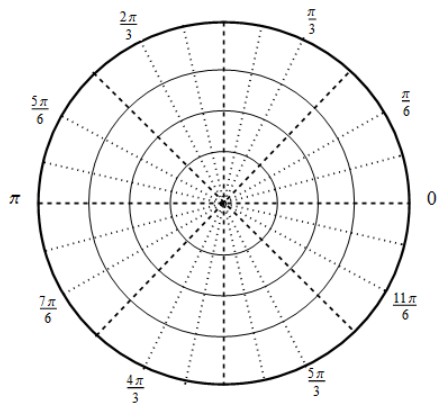
2. Sketch the graph. $r = 2 + 2\sin 2\theta$.



3. Sketch the graph $r = 2 - 2\sin\theta$



4. Sketch the graph $r = 2 + 3\cos\theta$



5. a) Find the Cartesian coordinates for the polar coordinate $(-3, 7\pi/4)$

b) Find polar coordinates for the Cartesian coordinate $(-2, 2\sqrt{3})$ where $r < 0, and\ \theta < 0$

c) Give two alternate versions on the polar point $(-2, -5\pi/4)$